

A Lightweight Empirical Comparison of Dreamer-Style and TD-MPC2-Style World Models for Continuous Control

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Abstract. Explicit and implicit world-model agents follow different representation-learning philosophies, yet their practical differences under limited compute remain underexplored. This paper presents an empirical comparison of two approaches for continuous control: a Dreamer-style recurrent state-space model with latent imagination and a TD-MPC2-style latent planning agent. Both agents are evaluated from low-dimensional state observations on DMControl tasks. The main matched benchmark is cartpole-swingup, where learning curves are compared over three random seeds using a shared interaction horizon and confidence intervals. A lower-budget walker-walk experiment is included as an additional extension. Beyond return curves, the study examines learned latent representations using PCA visualizations and linear probes for reward and task-state variables. In the reported setup, the TD-MPC2-style agent achieves substantially higher returns, while the Dreamer-style agent shows lower cross-seed variability on the main task. The latent analysis suggests that TD-MPC2 representations are more directly aligned with reward and key state variables on the extracted evaluation trajectories, whereas Dreamer representations are broader and less linearly accessible. The study is not intended as a capacity-matched benchmark, but as a reproducible comparison of performance, stability, and representation structure under a practical experimental regime.

Keywords: World models, model-based reinforcement learning, continuous control, Dreamer, TD-MPC2, latent representations, DeepMind Control Suite, linear probes.

1 Introduction

World models are a central idea in model-based reinforcement learning. Instead of learning behavior only from direct environment interaction, an agent learns an internal representation of the environment that can support prediction, planning, or policy optimization. The original World Models formulation showed that compact latent dynamics can be used for control, motivating a broad line of work on learned internal models for decision making [1].

Recent world-model agents differ in the way they use and shape latent representations. One line of work follows an explicit latent-imagination philosophy: Dreamer-

style methods learn recurrent state-space models and train actor-critic components from imagined trajectories in latent space [2, 3]. Another line follows a more implicit, control-oriented philosophy: TD-MPC and TD-MPC2 learn latent dynamics together with temporal-difference objectives, consistency learning, value estimation, and model-predictive planning [4, 5]. In this case, the latent representation is optimized less for full observation reconstruction and more for effective action selection.

Although these methods have achieved strong results on continuous-control benchmarks, their behavior under limited compute remains practically important. In small research projects, early-stage algorithmic studies, and reproducibility-oriented experiments, the relevant question is not only which method obtains the highest final return. It is also important to compare learning stability across seeds, training cost, and the structure of the learned latent space. The DeepMind Control Suite provides a suitable benchmark family for such analysis because it contains standardized continuous-control tasks with well-defined state and reward signals [6].

This paper presents an empirical comparison of two world-model implementations: a Dreamer-style recurrent state-space model with latent imagination and a TD-MPC2-style latent planning agent. Both agents are evaluated from low-dimensional state observations. The main matched benchmark is cartpole-swingup, where both methods are compared over three random seeds using a shared interaction horizon. A lower-budget walker-walk experiment is included as an additional extension rather than as a fully matched benchmark. The contribution of the paper is threefold: first, we compare the learning behavior of the two agents under a limited-compute setup; second, we analyze cross-seed stability using confidence intervals; and third, we examine learned latent representations using PCA visualizations and linear probes for reward and task-state variables. The paper is positioned as an empirical comparison and representation analysis, not as a new algorithmic contribution.

2 Problem Statement

The goal of this work is to compare two existing world-model paradigms under a reproducible continuous-control setup, rather than to propose a new algorithm. The first paradigm is represented by a Dreamer-style recurrent state-space model that uses latent imagination for behavior learning. The second is represented by a TD-MPC2-style agent that combines learned latent dynamics, temporal-difference objectives, and latent-space planning. The central question is how these approaches differ in learning behavior, cross-seed stability, and latent representation structure under a limited experimental budget.

The comparison uses low-dimensional state observations from the DeepMind Control Suite in order to reduce the influence of visual encoding and focus on latent modeling and control. The main matched benchmark is cartpole-swingup, where both agents are evaluated over three random seeds and compared on a shared interaction horizon. A walker-walk experiment is included as a lower-budget extension, not as a fully matched benchmark. Such compute and implementation constraints are common

in academic and small-lab research settings, where full-scale reproduction of state-of-the-art agents is often impractical.

The study is not capacity-matched: the TD-MPC2-style configuration used on the main task contains substantially more trainable parameters than the Dreamer-style configuration. Therefore, the results should not be interpreted as a definitive ranking of the two algorithms. Instead, the paper focuses on practical empirical trends: return curves, variability across seeds, and the linear accessibility of task-relevant information in learned latent states.

3 Related Work

Model-based reinforcement learning studies agents that use learned environment models to support decision making. Instead of relying only on direct trial-and-error interaction, these agents learn representations that can be used for prediction, planning, or policy optimization [7]. Latent world models are especially important in this area because they represent environment dynamics in compact internal states rather than directly in raw observation space.

One line of work follows an explicit latent-imagination philosophy. PlaNet introduced latent dynamics planning from pixels using a recurrent state-space model, showing that planning can be performed in a learned compact representation [8]. Dreamer extended this idea by learning behavior from imagined latent trajectories with an actor-critic objective inside the learned model [2]. DreamerV2 further developed this direction using discrete latent variables and demonstrated strong results on Atari tasks [9], while DreamerV3 proposed a more robust recipe across diverse domains with less task-specific tuning [3]. These methods place substantial pressure on the latent state to preserve predictive information that is useful for imagined rollouts.

A second line of work emphasizes control-oriented latent representations and planning. TD-MPC combines temporal-difference learning with model-predictive control in a learned latent space [4]. TD-MPC2 extends this approach toward scalable and robust continuous control by combining latent dynamics, consistency learning, value estimation, and planning with a policy prior [5]. These two philosophies place different pressures on the latent space: Dreamer-style agents organize representations for latent imagination, whereas TD-MPC-style agents organize them more directly around value prediction and action selection.

This difference motivates representation-level analysis. Return curves show whether an agent learns successful behavior, but they do not fully explain what information is preserved in the learned latent state. Diagnostic tools such as PCA visualizations and linear probes can provide additional evidence about whether reward and task-state variables are easily accessible from the representation. In this paper, these tools are used not as a complete theory of representation quality, but as practical diagnostics for comparing Dreamer-style and TD-MPC2-style latent spaces under the same experimental setting.

4 Methods

This section describes the two world-model agents compared in the study. The goal is not to modify the original algorithms, but to use representative implementations of two approaches to latent control: Dreamer-style latent imagination and TD-MPC2-style latent planning. Both agents are used with minimal task-specific modification, and the comparison focuses on their behavior under the same state-based continuous-control setting.

4.1 Dreamer-Style Latent Imagination Agent

The first agent follows the Dreamer family of world models [2, 3]. Dreamer-style agents learn a recurrent state-space model that summarizes past observations and actions in a latent state. The latent state contains a deterministic recurrent component for temporal memory and a stochastic component for uncertainty. This structure allows the model to predict future latent states and rewards while preserving information from previous interactions.

The learned model is used for latent imagination. Starting from states observed in replay, the agent generates imagined trajectories in latent space and trains actor-critic components on these trajectories. Policy learning is therefore supported not only by direct environment transitions, but also by rollouts produced by the learned world model. This is the main reason Dreamer-style methods are treated in this paper as representatives of the latent-imagination approach.

In this study, the Dreamer-style agent is evaluated from low-dimensional state observations rather than images. The experiment therefore does not test visual representation learning or pixel reconstruction. Instead, it focuses on whether the recurrent latent model can support behavior learning, reward prediction, and representation formation in a state-based continuous-control setting.

4.2 TD-MPC2-Style Latent Planning Agent

The second agent follows the TD-MPC2 approach [5], which extends the earlier TD-MPC framework [4]. TD-MPC-style agents also learn latent dynamics, but the latent space is organized more directly around control. The model is trained using temporal-difference objectives, consistency learning, reward prediction, value estimation, and a policy prior.

Action selection is performed through short-horizon model-predictive planning in latent space. Candidate action sequences are evaluated using the learned latent dynamics and value estimates, and the first action of the selected sequence is executed in the environment. This makes the learned representation directly useful for planning and return estimation.

This design places different pressure on the representation than Dreamer-style latent imagination. Instead of requiring the latent state to support imagined actor-critic rollouts in the Dreamer sense, TD-MPC2 trains the representation to be useful for value prediction and model-predictive control. As with the Dreamer-style agent, the

TD-MPC2-style agent is evaluated from state observations, keeping the comparison focused on latent control rather than image encoding.

4.3 Comparison Protocol

The comparison is organized around three questions. First, how quickly and reliably do the two agents improve their returns under a shared interaction budget? Second, how stable are the methods across random seeds? Third, what task-relevant information is linearly accessible from the learned latent states?

To answer the first two questions, the agents are compared using learning curves, final returns, best mean returns, and confidence intervals across seeds. To answer the third question, latent states are extracted from trained agents and analyzed using principal component analysis and linear probes. The probes are used to predict reward and selected task-state variables from the learned latent representation.

These diagnostics are not intended to provide a complete theory of representation quality. They are used as practical tools for comparing how the two methods organize task-relevant information internally under the same experimental setting. Details about seeds, interaction horizons, parameter counts, logging, and aggregation procedures are reported in the experimental setup.

5 Experimental Setup

The experiments use continuous-control tasks from the DeepMind Control Suite. All agents are trained and evaluated from low-dimensional state observations rather than image observations. This choice keeps the comparison focused on latent dynamics, planning, and control, instead of visual feature learning. The experiments were run on a GPU-equipped laptop with an NVIDIA GeForce RTX 3070 Laptop GPU with 8 GB of memory. The project adapts existing open-source implementations of DreamerV3 and TD-MPC2 rather than re-implementing both agents from scratch.

5.1 Tasks and Observation Modality

The main matched benchmark is cartpole-swingup. This task is used for the most direct comparison because both agents are evaluated over three random seeds and analyzed on a shared interaction horizon. The second task is walker-walk, which is included as a lower-budget extension to test whether the qualitative trends observed on cartpole-swingup persist in a more complex locomotion environment.

The walker-walk experiment is not treated as a fully matched benchmark because the TD-MPC2-style agent uses a smaller configuration and the shared comparison horizon is shorter. Therefore, the main empirical claims of the paper are based primarily on cartpole-swingup.

5.2 Training Configurations

The Dreamer-style agent is trained in a JAX-based setup using the `dmc_proprio` configuration. Both cartpole-swingup and walker-walk are trained with four parallel environments, a training ratio of 8, periodic logging, and periodic checkpointing. The Dreamer environment uses Python 3.12.3. The cartpole configuration contains 637,446 trainable parameters. Training metrics are saved to `metrics.jsonl`, episode scores are saved to `scores.jsonl`, and checkpoints are saved in a dedicated `ckpt` directory.

The TD-MPC2-style agent is trained using the provided Docker-based implementation with GPU acceleration. The TD-MPC2 environment uses Python 3.9.23 inside the Docker image `tdmpc2:1.0.1`. For the main cartpole-swingup comparison, the single-task configuration uses model size 5, corresponding to approximately 4,932,704 trainable parameters in the reported run. For walker-walk, TD-MPC2 uses model size 1 in order to keep the experiment feasible under the available compute budget. The TD-MPC2 training wrapper evaluates the agent every 10,000 environment steps. Evaluation rewards are saved to `eval.csv`, and the final model is saved under the `models` directory.

The comparison is not capacity-matched. The TD-MPC2-style configuration used on cartpole-swingup has substantially more trainable parameters than the Dreamer-style configuration. This difference is reported explicitly because it may affect performance, optimization behavior, and representation quality.

5.3 Evaluation and Logging

For both tasks and both methods, the random seeds are 1, 2, and 3. For cartpole-swingup, the intended interaction budget was 200,000 environment steps per seed. In the completed TD-MPC2 cartpole runs, however, the seeds ended at different horizons: 400k, 200k, and 150k environment steps. To preserve comparability, all TD-MPC2 multi-seed aggregates are computed only on the shared 150k horizon, and all direct Dreamer versus TD-MPC2 plots are truncated to the same range.

For walker-walk, the nominal interaction budget was 100,000 environment steps per seed. The comparison is reported on the shared 95k horizon because the Dreamer walker logs reached 96,096 environment steps and the plotting grid was aligned to 5k increments. Some TD-MPC2 logs extend beyond this horizon, but all walker-walk comparisons are truncated to the shared 95k grid. The walker-walk results are therefore interpreted as a preliminary extension rather than as a definitive comparison.

Performance is measured using return as a function of environment interaction steps. For plotting, all curves are interpolated onto a shared 5k-step grid. The cartpole-swingup grid spans 5k to 150k environment steps, while the walker-walk grid spans 5k to 95k environment steps. Dreamer curves are computed from episode scores stored in `scores.jsonl` and smoothed with a moving average window of 20 before interpolation. TD-MPC2 curves are computed from evaluation rewards stored in `eval.csv`. Because the two implementations expose different native logging pipelines, the reported values should be interpreted as implementation-level performance traces

aligned on a common interaction grid rather than as outputs of a single unified evaluator. For multi-seed plots, the mean and 95% confidence interval are computed across seeds at each grid point. These confidence intervals are used as descriptive uncertainty bands, not as formal hypothesis tests. Final mean return and best mean return on the shared grid are also reported.

The use of a shared interpolation grid is important because the two implementations do not produce evaluation records at identical step values. Aligning curves on the same grid makes the multi-seed summaries comparable while preserving the original logged trajectories. Therefore, the reported final and best mean values should be read as summaries on the shared comparison grid rather than as manually selected points from individual runs.

5.4 Latent Representation Analysis

To analyze learned representations, latent states are extracted from trained agents on cartpole-swingup. Principal component analysis is used to visualize the geometry of the latent space. The projected latent states are examined with respect to reward and selected task-state variables.

In addition, linear probes are trained to predict reward, pole angle, and cart position from the extracted latent states. The probes are used as diagnostic tools for measuring how directly task-relevant information is accessible from each representation. Because the probe results depend on the trajectory distribution used for extraction, they are interpreted as evidence about the analyzed evaluation trajectories rather than as a complete characterization of the full environment state space. The reported latent-space analysis is based on representative seed-2 checkpoints using the project script `latent_analysis_cartpole_seed2.py`. Therefore, this analysis should be viewed as an exploratory representation diagnostic rather than as a full multi-seed statistical comparison of latent spaces.

6 Results

This section reports the empirical results of the comparison. The main analysis is based on cartpole-swingup, which is the strongest matched benchmark in the study. The walker-walk task is reported as a preliminary lower-budget extension. Because the two implementations use different native logging pipelines, the results are interpreted as aligned implementation-level performance traces rather than as outputs of a single unified evaluator.

6.1 Main Matched Benchmark: Cartpole-Swingup

Figure 1 shows the main comparison on cartpole-swingup over the shared 150k interaction horizon. Both agents learn useful behavior, but the TD-MPC2-style agent reaches a substantially higher return regime in this setup. At 150k environment steps, the Dreamer-style agent reaches a final mean score of 115.30, whereas the TD-

MPC2-style agent reaches a final mean evaluation reward of 682.05. The best mean value on the shared grid is 115.30 for Dreamer and 775.74 for TD-MPC2.

These results indicate a strong empirical advantage for the TD-MPC2-style latent planning agent on the main benchmark. However, this advantage should be interpreted together with the experimental constraints described above. In particular, the comparison is not capacity-matched: the TD-MPC2 cartpole configuration contains substantially more trainable parameters than the Dreamer-style configuration. Therefore, the result should be understood as a comparison of the reported practical configurations, not as a definitive algorithmic ranking.

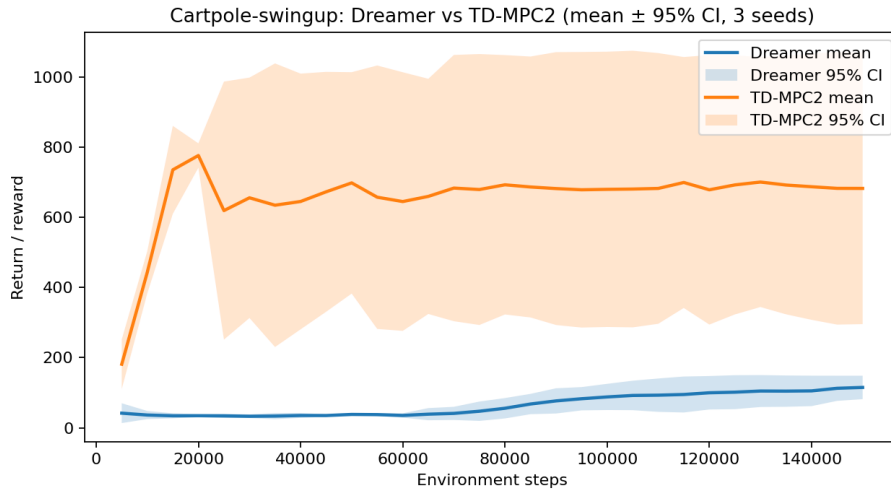


Fig. 1. Cartpole-swingup comparison on the shared 150k horizon. Curves show the mean over three seeds, and shaded regions show 95% confidence intervals. All curves are interpolated onto the same 5k-step grid.

Overall, TD-MPC2 reaches higher returns under the reported configurations, while the Dreamer-style agent remains more stable across seeds on the main benchmark.

6.2 Cross-Seed Stability

The confidence intervals show that higher mean performance does not necessarily imply higher cross-seed stability. On cartpole-swingup, the Dreamer-style agent has a final 95% confidence-interval half-width of 33.28 at 150k steps. The TD-MPC2-style agent has a much larger half-width of 386.47 at the same horizon.

This difference reflects the seed-level behavior observed in the runs. Dreamer remains in a lower-return regime, but its three seeds are closer to one another. TD-MPC2 reaches much higher returns on average, but its seed variability is larger. Thus, the main cartpole result is not simply that TD-MPC2 performs better. Rather, the result shows a trade-off: TD-MPC2 achieves stronger mean performance under the reported setup, while Dreamer shows lower variability across seeds.

Table 1. Quantitative summary of return curves on the shared comparison grids.

Task	Method	Horizon	Final mean	95% CI half-width	Best mean
cartpole-swingup	Dreamer-style	150k	115.30	33.28	115.30
cartpole-swingup	TD-MPC2-style	150k	682.05	386.47	775.74
walker-walk	Dreamer-style	95k	38.64	4.01	38.70
walker-walk	TD-MPC2-style	95k	972.46	5.36	972.46

The confidence intervals in Table 1 are descriptive uncertainty bands computed across three seeds. They are not used as formal hypothesis tests.

6.3 Latent-Space Analysis

To examine whether the two agents organize task-relevant information differently, latent states were extracted on cartpole-swingup and analyzed using linear probes and PCA visualizations. Table 2 summarizes the linear-probe results, while Figure 2 provides a qualitative PCA view of the same latent representations.

Table 2. Linear-probe results on cartpole-swingup latent representations.

Method	Target	R ²	MAE
Dreamer-style	Reward	0.7666	0.0341
Dreamer-style	Pole angle	0.4715	1.3647
Dreamer-style	Cart position	0.7993	0.2831
TD-MPC2-style	Reward	1.0000	0.0007
TD-MPC2-style	Pole angle	0.9429	0.0496
TD-MPC2-style	Cart position	1.0000	0.0000

For the Dreamer-style agent, reward and cart position are partly linearly accessible from the latent state, with R² values of 0.7666 and 0.7993, respectively. Pole angle is less directly accessible, with an R² value of 0.4715. For the TD-MPC2-style agent, the probes obtain much higher values on the same target variables, including rounded R² values of 1.0000 for reward and cart position and 0.9429 for pole angle.

The PCA visualization complements these numerical results. Dreamer’s latent space appears broader and less structured, while TD-MPC2 forms a thinner, more organized manifold on the analyzed trajectories. Together, the probe and PCA results suggest that, on the extracted evaluation trajectories, the TD-MPC2-style latent representation is more directly aligned with reward and key task-state variables.

This conclusion should be interpreted cautiously. The analysis is based on representative seed-2 checkpoints and does not constitute a full multi-seed statistical comparison of latent spaces. In addition, very high probe scores may partly reflect the structure of the evaluation trajectories rather than general linear accessibility over the full environment state distribution.

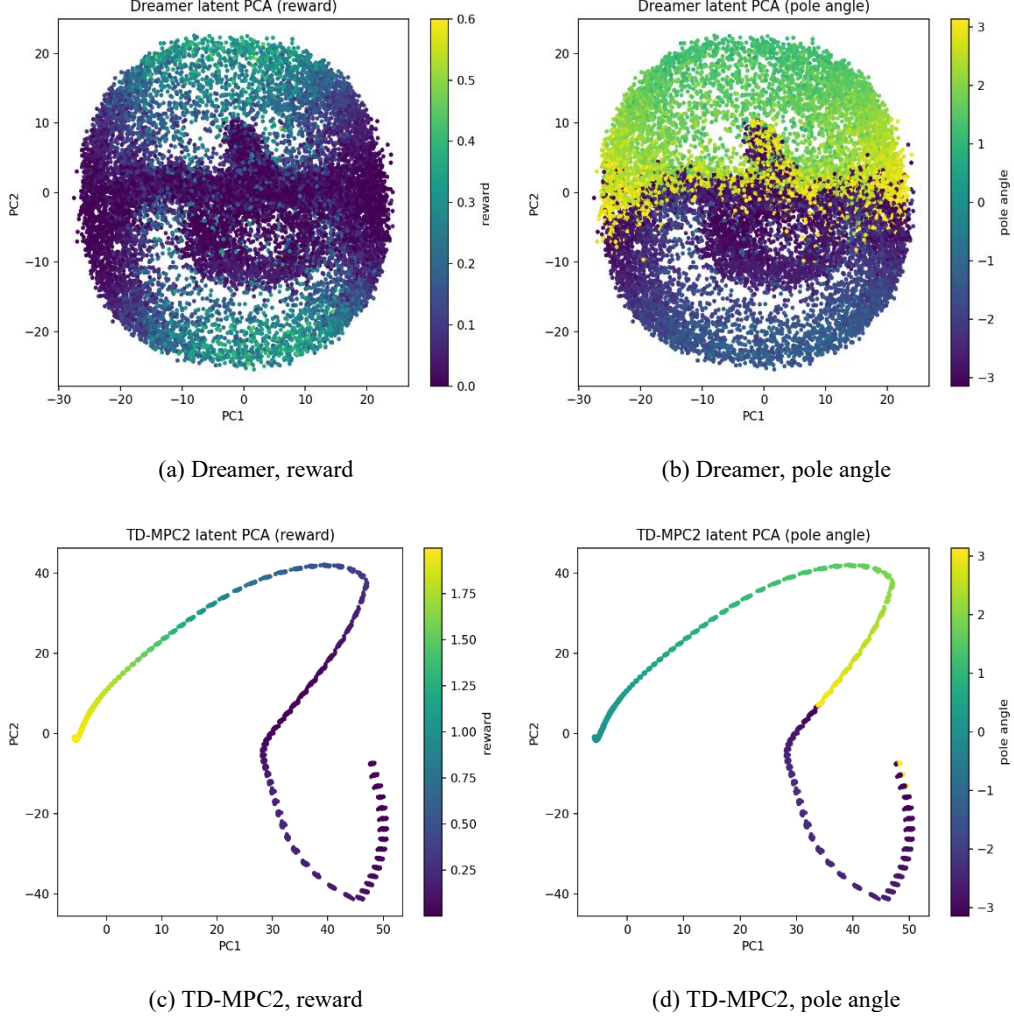


Fig. 2. PCA visualization of cartpole-swingup latent states. Panels show (a) Dreamer latents colored by reward, (b) Dreamer latents colored by pole angle, (c) TD-MPC2 latents colored by reward, and (d) TD-MPC2 latents colored by pole angle. Dreamer’s latent space appears broader and less structured, while TD-MPC2 forms a thinner, more organized manifold on the analyzed trajectories.

6.4 Preliminary Lower-Budget Extension: Walker-Walk

Figure 3 reports the walker-walk extension on the shared 95k horizon. This task is more complex than cartpole-swingup, but the comparison is less strictly matched because TD-MPC2 uses model size 1 and the experiment is run under a shorter shared horizon. Therefore, the walker result is treated as a preliminary extension.

The qualitative trend is consistent with the main benchmark. At 95k environment steps, the Dreamer-style agent reaches a final mean score of 38.64, while the TD-MPC2-style agent reaches a final mean evaluation reward of 972.46. The corresponding 95% confidence-interval half-widths are 4.01 for Dreamer and 5.36 for TD-MPC2. Unlike cartpole-swingup, where TD-MPC2 shows high seed variability, the walker-walk extension shows both high mean performance and a tight uncertainty band for TD-MPC2.

This result supports the observation that the TD-MPC2-style agent is stronger under the reported practical configurations. However, because the walker experiment is lower-budget and not fully matched, it is used only to support the qualitative trend observed on cartpole-swingup, not as an independent definitive benchmark.

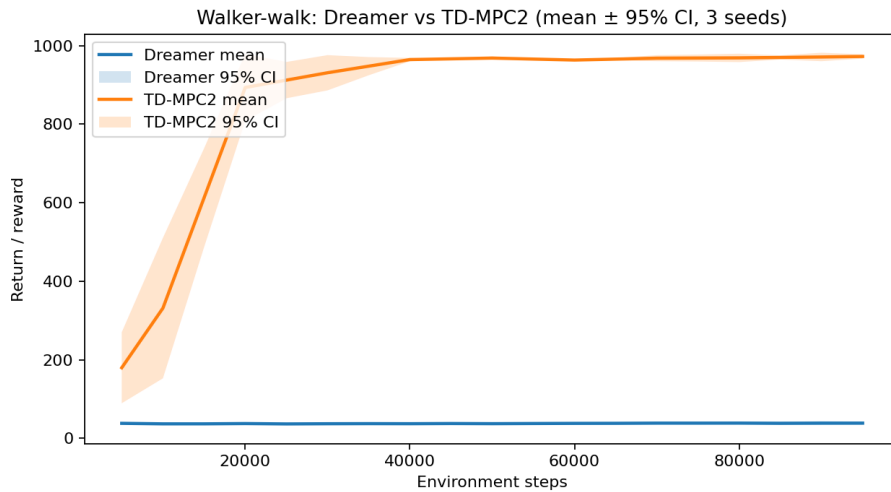


Fig. 3. Walker-walk comparison on the shared 95k horizon. Curves show the mean over three seeds, and shaded regions show 95% confidence intervals. The task is treated as a preliminary lower-budget extension.

Together, these results provide a practical comparison of the two world-model approaches under limited compute, highlighting differences in return, seed stability, and latent representation structure.

7 Limitations

The results should be interpreted in light of several methodological and practical limitations.

First, the comparison is not capacity-matched. The TD-MPC2-style agent uses a substantially larger configuration than the Dreamer-style agent on the main cartpole-swingup benchmark. Therefore, the observed performance differences may reflect not only algorithmic differences, but also differences in model capacity, optimization defaults, and implementation maturity.

Second, the experiments are performed under a limited compute budget on state-based observations. This design makes the comparison practical and reproducible, but it does not represent a full-scale reproduction of the best reported DreamerV3 or TD-MPC2 results. In particular, the Dreamer-style results should not be interpreted as optimal Dreamer performance.

Third, the study focuses on two DeepMind Control Suite tasks rather than a broad benchmark suite. This limits the generality of the conclusions, especially for more complex domains or pixel-based settings.

Fourth, the two implementations expose different native logging and evaluation pipelines. The reported curves are aligned on a common interaction grid, but they are not produced by a single unified evaluator. For this reason, the results are best interpreted as implementation-level performance traces under matched tasks and shared horizons.

Fifth, the walker-walk experiment is a lower-budget extension rather than a fully matched benchmark. It supports the qualitative trend observed on cartpole-swingup, but the main empirical claims of the paper are based primarily on the cartpole-swingup comparison.

Finally, the latent-space analysis is exploratory. It is based on representative seed-2 checkpoints and extracted evaluation trajectories, rather than on a full multi-seed statistical analysis of latent representations. The linear-probe results therefore show how accessible task-relevant variables are on the analyzed trajectories, but they do not fully characterize the complete environment state distribution.

These limitations motivate future work with capacity-matched models, unified evaluation pipelines, larger seed counts, broader DeepMind Control Suite coverage, and more rigorous latent-probe generalization tests.

8 Conclusion

This study provides a practical comparison of two world-modeling approaches for state-based continuous control: a Dreamer-style latent imagination agent and a TD-MPC2-style latent planning agent.

On the main cartpole-swingup benchmark, the TD-MPC2-style agent achieved higher mean returns under the shared comparison horizon, while the Dreamer-style agent showed lower cross-seed variability. Latent-space diagnostics indicated that TD-MPC2 representations were more directly aligned with reward and task-state variables on the analyzed trajectories, whereas Dreamer representations were broader and less linearly accessible.

The walker-walk extension showed a similar qualitative performance trend in favor of the TD-MPC2-style agent, but this result was treated as preliminary because the comparison was lower-budget and not fully matched. Overall, the findings should be interpreted within the limitations of the study: the comparison is not capacity-matched, uses a limited number of tasks and seeds, and relies on implementation-specific logging pipelines.

Together, these findings highlight the importance of evaluating world-model agents not only by return, but also by stability, implementation constraints, and representational structure. Future work should extend the analysis to capacity-matched configurations, broader DeepMind Control Suite coverage, unified evaluation protocols, and more rigorous multi-seed latent-representation analysis.

Code Availability

The code, plotting scripts, numerical summaries, and selected representative checkpoints used in this study are available at: <https://github.com/Khaled-Deeb/world-models-explicit-vs-implicit>. The repository includes reproduction instructions and version information for the adapted DreamerV3 and TD-MPC2 implementations.

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